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How Raven Song Flutes Are Made

Introduction and History

This booklet is intended to give the reader insight on how I make my Raven Song Flutes. It is not directly intended to be a “how to” book, however the reader may, by following the process, use it for the purpose of learning how to make a flute. In doing so, remember that my methods are what work for me. You may choose alternatives depending upon your experience with woodworking, tools available and your own personal choice. My shop is designed and set up for the primary purpose of making flutes and I have a considerable amount of equipment. I love electricity and tools, and love to make jigs to help in the process. When I started making flutes back in 1993, my first flutes were made almost completely by hand. Crude as they were, they worked, but I also know that making a good sounding, well made flute by hand takes a lot of time. Since becoming a flute maker I have made and passed on to their caretakers over 1800 flutes. That would not have been possible without the use of modern tools and equipment.

You might be interested in how I came to be a flute maker. I believe from a spiritual aspect, I was chosen. Back in 1993, I had a chimney sweep business and was doing quite well with it, however, I was not really satisfied. One day, while moving a small table in preparation to cleaning a fireplace, a local newspaper fell to the floor. In picking it up, I noticed an article about a flute making workshop that was to be held that very evening in the next town. Having always had an interest in Native crafts, and always having loved the sound of the flute, I decided to attend. The presenter was an elderly Native American named Hollis Littlecreek. Grandfather Hollis was a fifth generation flute maker from the Martin Clan of the Ojibway people. Hollis showed his flutes that he had in various stages of completion and said he would tell us anything we wished to know about making them with one exception. The exception was his formula, which he said was to be passed on to his grandson. That sounded fair enough to me and I spent the evening engrossed in learning all I could.

Upon returning home, I began making my first flute and then my second and third. I still have my very first flute and it brings a smile to my face each time I look at it. It's pretty crude, but it did make a sound, and without it, I would not be making the flutes I make today. The next time Hollis was in the area, I went to meet with him and showed him my creations. This continued for the next few years. Each time I met with him he would answer a few questions and offer bits of advice, and then I knew that the session was over.

At the time, Hollis was living in Sedona, Arizona and I had the opportunity to visit him at his home. Almost immediately upon arriving, he asked if I had brought any flutes with me. Saying that I had, he said, “well don't just stand there, go get them”. He looked them over and sat back in his usual fashion and then said, “If you keep this up, I'm going to be out of business. Now let me show you something”. He then proceeded to give me his formula and I realized that what he had said at the first workshop, did not actually mean his biological grandson. He was looking for those who wanted it bad enough to stick with it. I guess I did, and I have since gone beyond. Hollis has since passed over to the Spirit World, however, there have been several occasions while making flutes that his presence has been felt looking over my shoulder. So to Hollis, I say a huge “megwitch” thank you in Ojibway.

Preparing the shop and wood selection.

Before beginning a new flute or batch of flutes as is sometimes the case, I like to spruce up the shop and perform a smudging ceremony and ask that all goes well with the new flutes. I also lay out the wood I am to use and actually ask each piece if it wishes to become a flute.

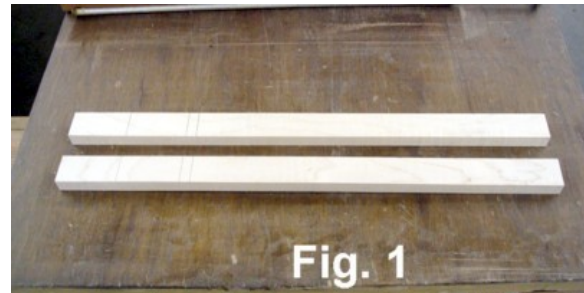
The response is usually in the affirmative, however there have been times when a piece of wood will respond by cracking, curling or twisting, or finding other ways to say it does not. Sometimes it will occur during one of the steps along the way. I can usually tell whether it is a mistake on my part, or a response from the wood.

I also only work on flutes while in a good, loving mood. If I am upset or in some other frame of mind, I fear the mood is passed onto the flute and will be reflected in its final sound or voice.

Now, let's get on to how I make my flutes.

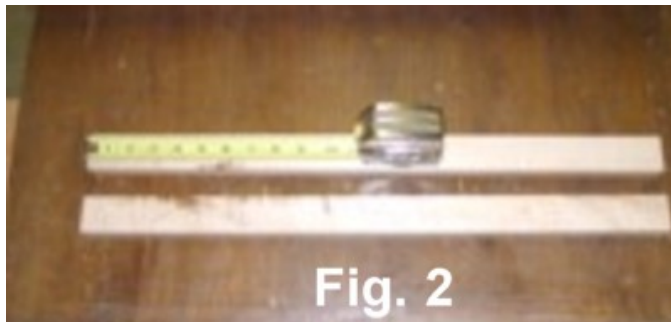
Cutting the blanks.

Raven Song Flutes are made in the "split" method where I begin with two halves, route out the center and glue the two pieces together. Whenever possible I like to begin with a solid piece of wood and slice it in half, then match the grain when it is glued back together. Depending on the wood I have, sometimes I cut two adjacent pieces from the same board. Most of my flutes begin with a 1½ inch square and I usually cut them 24 inches in length. This allows me the greatest latitude in keying and simplifies set up. Figure 1 shows two 1½ X ¾ X 24 inch pieces of Soft Maple that will create a 1½ X 1½ blank.



Laying out the blanks.

The next step is to lay out the blanks for routing the main bore and the slow air chamber, hereafter referred to as the SAC. Figures 2 and 3 show this simple process. Most all of Raven Song Flutes are laid out in the same manner. I mark off the blanks at 3 inch, 6 inch and 6½ inch intervals and using a square, I draw lines on what will be the inside of the flute. These lines are then used as a guide when routing the bore, SAC and the blow hole.



Carving or routing the bore, SAC and blow hole.

The next step in the process is to carve or route the bore, SAC and blow hole. This is one of the most critical steps as it is very important to make the bore as smooth and consistent as possible. It is possible to sand the bore to get it smooth, however, if the wall thickness varies, especially in the area where the finger holes will be, the flute will become difficult to tune properly. I love electricity and jigs, therefore, I use a 3½ horsepower router and have built a rather elaborate jig for this process. Figure 4 shows the router and jig and Figure 4A shows the completed blank. Figure 4B shows two hand tools that can also be used to complete the process. If you are interested in making flutes and plan to make several, I recommend investing in a good router that will take ½ inch shank bits. Routing out the blanks is much quicker than carving them by hand and the end result is much more consistent.

**Fig. 4**

The jig I use is based on the principle of diagonal pins which automatically centers the cutter regardless of the width of the blank. This is very handy and all of the routing operations that require centering can be done on this jig. I do not have plans for the jig available, however a little thought and ingenuity and you can come up with your own version. It is well worth the effort if you are planning on making a lot of flutes. For someone looking to make one or just an occasional flute, the hand tools will work fine.

**Fig. 4A****Fig 4B**

De-burring the blanks.

The next process is to de-burr the edges of the routed areas on the blanks. This is best done by hand with a piece of fine sandpaper, being careful not to round over the edges. This step is more critical than it appears as the burrs, regardless of how small will interfere when the blanks are glued up. Figure 5 shows this process.

**Fig. 5**

Laying out the True Sound Hole and the SAC Hole.

This next process and the ones that follow will create the sound mechanism and will decide how the flute will sound. Great care should be taken during the next few steps.

First of all, from the two blanks, decide which one will be the top of the flute. The decision may be based on the grain configuration, wood color or some other pleasing feature. There may be no difference between the two pieces and either one will work fine.

Next, at the partition between the main bore and the SAC, (see Figure 6), at the point where the main bore curves up to the partition, make two marks approximately $\frac{3}{16}$ inch apart on the side wall. Do the same at the SAC. Next, using a square, transfer these marks to the outside of the top blank. Figures 6, 7 and 8 show this process.

Next, divide the bore diameter in half, $\frac{3}{8}$ inch for a $\frac{3}{4}$ inch bore; $\frac{7}{16}$ inch for a $\frac{7}{8}$ inch bore and $\frac{1}{2}$ inch for a 1 inch bore. Now, subtract that from the width of your blank and divide that in half. For example, a $\frac{7}{8}$ inch bore would be $\frac{7}{16}$ inches. If your blank width is $1\frac{1}{2}$ inches, then the resulting difference would be 1 and $\frac{1}{16}$ inches. Dividing that in half would be $\frac{17}{32}$ inches. Next, using a tri-square, set the $\frac{17}{32}$ on the blade and allow a little for pencil thickness. Next, draw lines perpendicular to the lines already on your blank from both sides of the blank. (see Figure 8A) This should give you two rectangles that are $\frac{7}{16}$ X $\frac{3}{16}$ approximately, one on either side of the partition between the main bore and the SAC. Great care

should be taken while laying out these areas as well as in the work to be done next, as this work will create the sound mechanism of the flute



Fig. 6



Fig. 7



Fig. 8



Fig. 8A

Milling the True Sound Hole.

Now the true sound hole (hereafter referred to as the TSH) and the hole from the SAC can be milled. I have a small mini-milling machine, however, the holes can be milled using a cross vice on a drill press or even by drilling a series of holes with a hand drill and filing out the holes later. I use the milling machine because of its' accuracy. If you are only making one flute or an occasional flute, you can use one of the other methods.

The slots should be milled $7/32$ of an inch and this is critical. If the slots are too large, you will loose both volume and tone. Figure 9 shows a flute blank in the milling machine ready to be milled. Figure 10 shows the milled blank. Note that the corners are round. They will be squared off in the next process.



Fig. 9

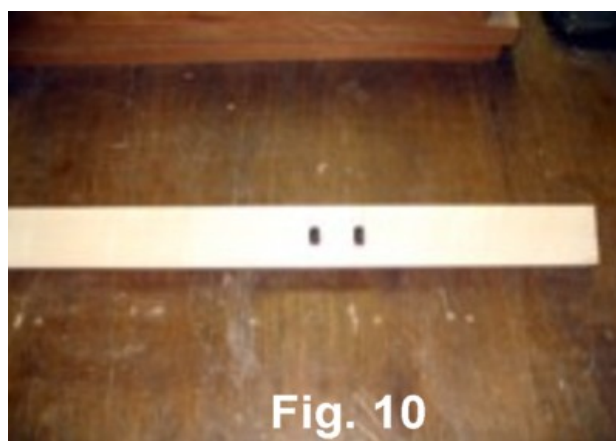


Fig. 10

Shaping the True Sound Hole and the Slow Air Chamber Hole.

The next process is perhaps the most critical area in making a good sounding flute. For this I use Pillar files because of their accuracy. I have three sets of these files, extra narrow, narrow and regular and I have a number 0, 00, and 2. A source for these files can be found in the appendix. Here again if you are only making a single flute you can pick up an inexpensive set of needle files at a Dollar Store that will serve you well.

To file the corners square I like to clamp the blank down on a flat surface and file vertically. The main thing is to go slow and make the corners square without making the hole larger than the 7/32 in length. You also want to maintain the width, which should be one half of the bore diameter. Figure 11 shows this process and Figure 11A shows the Pillar Files.

While shaping the TSH and the SAC holes, care should be taken to make the interior walls square as well. Figure 12 shows the blank with the hole corners filed square.



Fig. 11



Fig. 11A

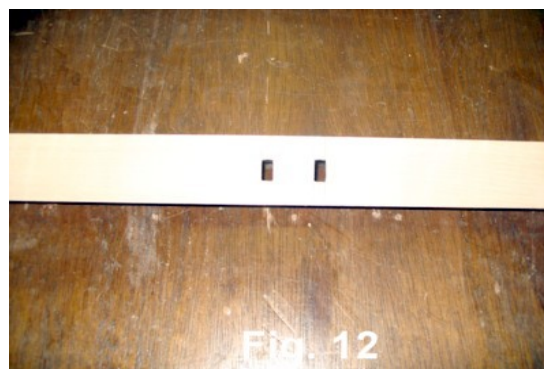


Fig. 12

Chiseling the ramp.

Once the TSH and the SAC hole are squared away, the next step is to turn the top blank over and chisel out the ramp on the forward edge of the TSH. This is another critical step, in fact, it is perhaps the most critical. For this process, I use a sharp chisel that is a little narrower than the TSH. Holding the chisel with the flat side down and at 45 degrees, I chisel out the ramp being careful to keep the chisel flat so that the final edge of the ramp is even in width. After I have chiseled the ramp, I cut away the excess on the ramp sides with a sharp knife, again being sure to keep the sides of the ramp at 90 degrees. Then, using my Pillar files, I file the sides and the ramp to assure all surfaces are clean, square and that the cutting edge at the base of the ramp is even. Figures 13 and 14 show this process.

The other critical thing is the thickness of the leading edge of the TSH. Since this is where the air is split and the sound created, it is essential that this edge be the same thickness all the way across and I try to make mine between 1/32 and 3/64 of an inch thick. Also, this edge does not want to be a knife edge. See Figure 14A. I also chisel the leading edge of the SAC hole to make the air flow nice and smooth. The top edge of this ramp does not have to be as thin as the TSH ramp.



Fig. 13



Fig. 14

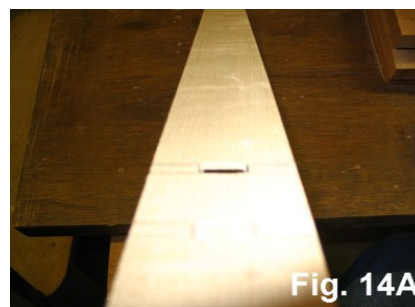


Fig. 14A

Sanding the inside of the bore and the SAC.

With the sound mechanism complete, the next step is to sand the inside of the bore and the SAC on both the upper and lower blanks. I do this by clamping each blank on my workbench and using my special blocks (see Figure 15) I sand both pieces to 600 grit. You can go finer if you desire but I find 600 to be a satisfactory limit. The main thing is to have the inside of the bore as smooth as possible. Sometimes it takes just a little sanding and other times, because of rough grain, it takes quite a bit of sanding. Figure 15A shows this process.



Fig. 15



Fig. 15A

Finishing the bore and SAC.

Once the bore is sanded to satisfaction, three thin coats of shellac are applied. The first coat dries almost immediately and can be sanded with 600 grit paper. The second and third coats usually take a bit longer to dry so I like to allow 20 to 30 minutes between coats and sanding. In addition to applying shellac to the bore and SAC, I also apply shellac to the sides of the TSH and the ramp, and also the SAC hole. To do this, I use disposable plastic brushes that can be found in the Dollar Store or a similar place. Pipe cleaners also work well for this. Figures 16 and 17 show this process.

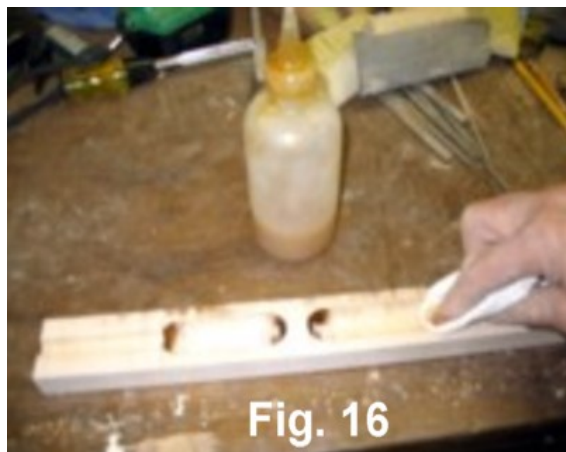


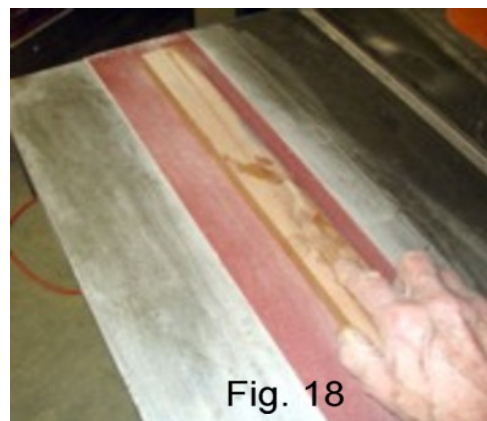
Fig. 16



Fig. 17

Sanding the blanks for flatness.

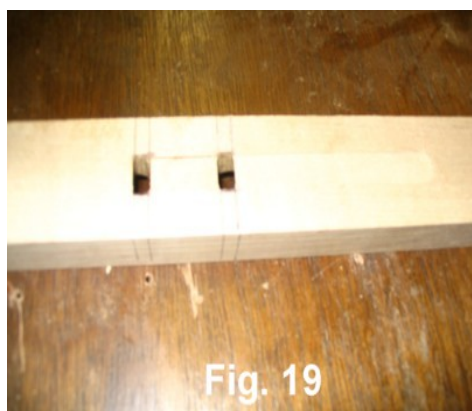
With the bore and SAC finished, I next check the blanks for flatness and to be sure I will get a good glue joint. To do this, I keep a piece of adhesive backed sandpaper on the table saw extension. This gives a true, flat surface to sand the blanks. Sanding the blanks also removes any excess shellac that might have spilled on the glue surfaces. Figure 18 shows this process.



Routing the airway.

The next step is to route the airway between the SAC hole and the TSH. It is critical that this area be absolutely flat and of consistent depth. It should also be only 1/32 of an inch deep. This depth is also critical. Too deep and you lose sound, not deep enough and you lose volume. I also route what I call the air run out beyond the TSH. This run out is one half the depth of the airway, or 1/64 of an inch thick. The purpose of this run out is to align the cutting edge of the TSH with the air coming from the SAC. It also makes the cutting edge of the TSH 1/64 thick which is what you want it to be. Figure 19 shows the blank with the airway and run out routed in.

I route both the airway and the run out on my routing jig which automatically centers them. I also have made several depth gauges which helps in setting up the router for the various operations. Figure 19A shows some of these gauges.



Routing the bird channels.

One feature of Raven Song Flutes that I feel is unique, is the way I mount the birds. Rather than just having the bird sit on the flat surface, I have two grooves routed into the flat area, one on either side of the TSH. The bird is routed out to form two ridges that correspond to the grooves routed into the flute body. This allows the bird to be moved back and forth for proper tuning while not allowing it to twist



Gluing the blanks together.

Now the blanks are ready to be glued together to make a whole flute blank. Several things are important during this process. First, glue must be applied to both blanks in a thin layer. Too much glue results in a sloppy situation and leaves excess glue in the joint. Figure 21 shows a proper amount of glue. The next thing to be sure of is that the bore and blow holes are properly lined up. If they are not you will have a difficult time getting them right once the glue has dried. I place a spring clamp on the side of both ends to help assure that the bore and blow hole are properly aligned. Even then, I check them several time while the glue is still wet. As more clamps are applied the two blanks can move around so I go slow and check my alignment often. I use spring clamps and use as many as necessary. Usually about ten clamps are sufficient, however, I will use more if the situation dictates. Figure 22 shows a typical glue up process.

Once all the clamps are in place, I swab the bore with a damp rag to remove excess glue from the inside of the bore. This is done by attaching a piece of rag to a long dowel that has a slot cut in the end. Then the rag is dampened and pushed into the bore. Usually, two or three passes is sufficient. I use a small pen light to shine down into the TSH and look up the bore to see if it is properly clear of excess glue. Once I am satisfied that the blanks are properly aligned and the bore is clear, I set the flute aside to dry for a minimum of two hours, but usually overnight. Figure 23 shows the swab I use to clean out excess glue. Figure 24 shows a glued up flute.



Fig. 21

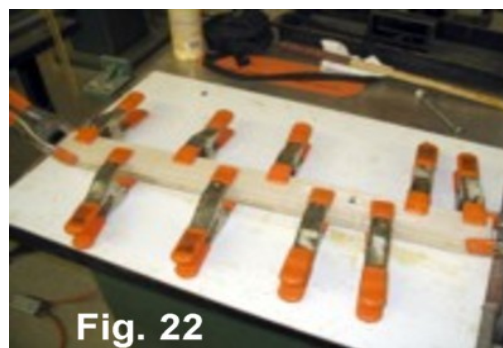


Fig. 22



Fig. 23



Fig. 24

Once the glue has dried I scrape the outside of the flute to remove excess glue and drill out the blow hole with a 3/8 inch drill bit to remove excess glue from the inside of the hole. See Figures 25, 25A



Fig. 25



Fig. 25A

Shaping the flute.

Now that I have a well glued blank I am ready to shape it on the lathe. The first thing I do is to trim the ends which can be a little off. I do this on a trim saw, however it can also be done by hand. For this I use a Japanese pull saw which works very well. Figures 26 and 26A show this process.

Once the flute has been trimmed, it is ready to be turned on the lathe. I have a special set of tools that were made specifically for turning flutes and I highly recommend anyone who is serious about making flutes to invest in a set. The source is listed in the appendix. Figure 27 shows the tools. They have made a tremendous difference in my flutes. The other tool that has helped in turning is a steady rest that I made. A rest is not always necessary, however, when a flute begins to vibrate harmonically, it is essential. The steady rest can be seen in Figure 31.

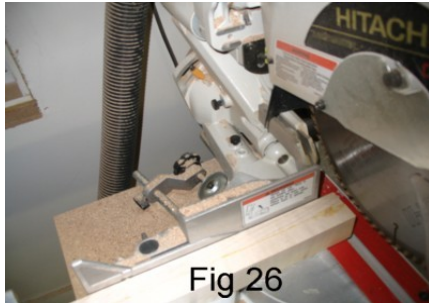


Fig. 26

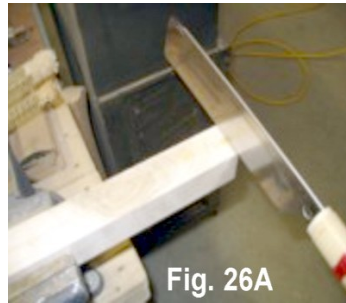


Fig. 26A



Fig. 27

The first step in turning is to mark out the area around the TSH to establish the flat area where the bird will sit. I mark out about a four inch area beginning $\frac{1}{2}$ inch in front of the TSH and running back toward the blowhole. I then draw lines using a square around the flute. See Figure 28. This area will not be turned. The next step is to divide this area into thirds, for example, for a $1\frac{1}{2} \times 1\frac{1}{2}$ blank, I set my tri-square at $\frac{1}{2}$ inch, allowing a little for the pencil mark and draw parallel lines perpendicular to the first line I drew around the flute. See figures 29 and 30. These lines will serve as a guide for hand shaping this area of the flute after turning. Next I begin turning the flute. I begin on the right hand side of the lathe and turn a section three to four inches. I turn this area to about whatever I want for a finished diameter. Normally I try to have my finished wall thickness to be $\frac{3}{16}$ of an inch, so the finished diameter will be the bore diameter plus $\frac{3}{8}$ of an inch. For a $\frac{7}{8}$ inch bore it would be $1\frac{1}{4}$ inches. See figures 31 and 32. I then continue turning the flute making several measurements as I go to assure I maintain a consistent wall thickness. Once the bore end has been turned, I turn the blow hole end making it compatible with the bore end and having a pleasing shape. See figure 33.

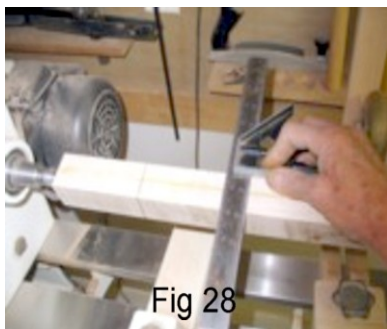


Fig. 28



Fig. 29

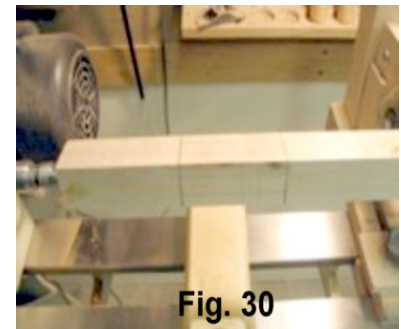


Fig. 30



Fig. 31



Fig. 32

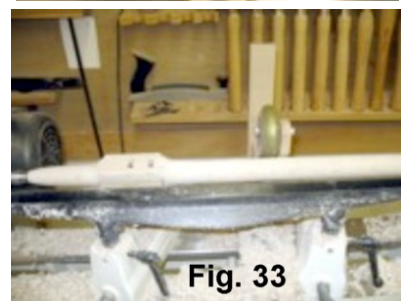


Fig. 33

Once the flute is turned, I shape the flat or nest area by hand. To do this I use a micro-plane or a small hand plane to shave the two corners on the bottom of the flute at 45 degrees and to the parallel lines I drew earlier. Because of the method I use to attach my birds, my flutes are a little wider at the nest area, therefore when shaping this area I adjust accordingly and shape it to have a pleasing look. See figure 34

and 35. Figure 36 and 37 show the stages of shaping the nest area. Figure 36 shows it after the corners are planed to 45 degrees and figure 37 shows it after the corners have been rounded giving the flute a finished appearance.

The next step is to sand the flute while still on the lathe. I sand my flute to 400 grit sandpaper at this point. They will receive additional sanding later on. While sanding on the lathe, I am careful not to sand the area around the nest as this would violate the integrity of the nest area. This area will be hand sanded later on. See figure 38.



Fig. 34



Fig. 35



Fig. 36

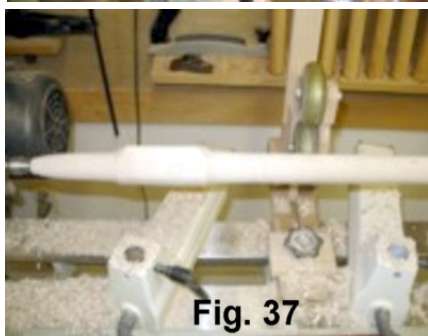


Fig. 37



Fig. 38

Figure 39 shows a flute completely turned and shaped ready to be tuned to its fundamental. Before tuning however, I further treat the SAC and blow hole with a “shellac bath”. I pour about a good tablespoon of shellac into the SAC at the SAC hole. Then, holding a finger over the blow hole, I turn the flute over and around to further coat the inside of the SAC. Then I pour out any excess shellac and set the flute aside to dry for at least an hour.

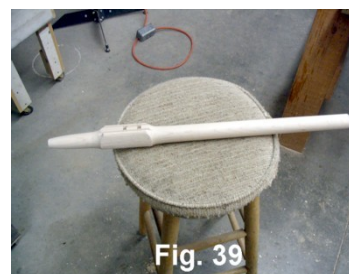


Fig. 39

Establishing the flute fundamental.

With the SAC dry, it is time to establish the fundamental tone or key of the flute. To do this, I simply hold a bird in the proper position on the nest and blow gently into the flute while checking my tuner. Sometimes I have a pre-determined key in mind and other times I take what the flute wants to give me. In either case, I establish the fundamental key by cutting the bore end shorter until the tuner reads in the center at the proper key. Sometimes I cut off as much as two inches, and other times maybe one quarter to one half an inch. I go slow with this process as I approach the fundamental and sneak up on it by cutting off very small pieces. Once I have the fundamental established, I will cut off just a little more to raise the fundamental, or sharpen it, since it will be lowered when I drill the finger holes. Quite often, even when I begin tuning with the fundamental on the sharp side, I will still have to cut off a little more once I have the flute completely tuned.

The reason for this is in a lot of sound physics occurring in the flutes main chamber. To simplify it, think of a six-hole flute as seven different flutes. The main bore being one, and each of the six finger holes being another tiny flute. Since each of the six tiny flutes adds volume to the flute, the fundamental will be flattened as each one is enlarged in the tuning process. That, is why it is usually necessary to cut more off the bore to re-establish the fundamental after the flute is tuned. Figures 40 and 41 show this process.



Fig. 40



Fig. 41

Laying out the playing holes.

With the fundamental established, I lay out the playing holes. There are many ways and all sorts of mathematical formulas to determine the proper location and size for the playing holes. To keep it simple, I use a very easy method and leave the mathematics to the physicists. First, using a straight edge and a soft pencil, I draw a line down the center of the bore beginning about two inches below the TSH. (See figure 42)

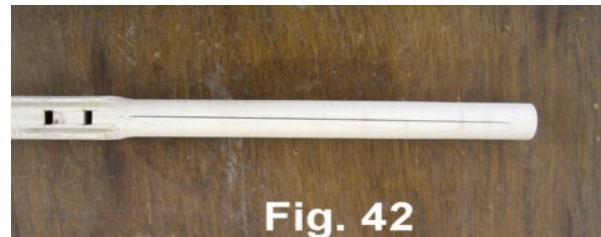


Fig. 42

Next, I measure from the bottom of the TSH to the end of the flute and mark the center point. Since this point will not be one of the hole locations, I mark the hole with a "C" for center so I do not get confused later on when I drill the playing holes. Next, from the mark, I measure down the flute approximately $\frac{3}{8}$ of an inch. I say approximately because on short flutes I usually only measure down $\frac{1}{4}$ of an inch, On longer flutes with a bore length of 15 inches or longer, I measure $\frac{3}{8}$. Next I mark that point as the location of the third hole from the bottom of the flute. Then I mark out the second and first holes equal distance apart. Now I have the bottom three holes located all equal distance apart. Usually I use $1 \frac{1}{8}$ inches for this distance, however if it is a short flute or I know that the person who will be playing the flute has small hands, I will make this distance $1 \frac{1}{16}$ of an inch. Next, I return to the center mark and mark the next three holes all the same distance as the bottom holes. Figures 43 and 44 show this process.

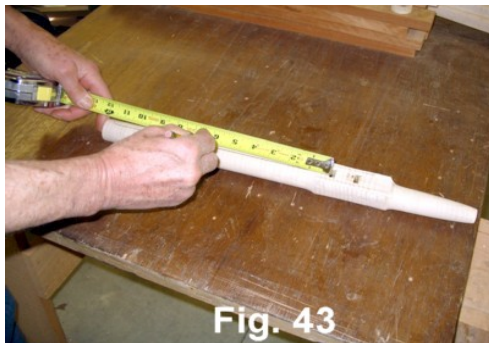


Fig. 43



Fig. 44

Drilling the playing holes.

Next, I drill the six playing holes. I drill these holes on a drill press, however if you do not have a drill press, you can drill them very carefully with a hand drill. I use a $\frac{1}{8}$ inch drill for these holes to be sure that the holes are smaller than they want to be when tuned. Figure 45 shows the drilling process.



Fig. 45

Tuning the flute.

The tuning process I use tunes flutes to Mode 1 and Mode 4 of the pentatonic scale. This is the usual tuning for this type of flute. The individual notes for each key are listed in the appendix. If you are looking for some other scale, I recommend searching the web for the information. It is possible to play a diatonic scale on these flutes and with a little practice you will find you can play just about any tune you desire. With the playing holes drilled, I re-check the fundamental and adjust it if necessary by trimming very small amounts away from the flute. Once the fundamental is correct, I begin with the lowest hole from my mouth. (this is hole number 1), and using burning sticks, I enlarge the hole in very small increments while continually checking the tone on the tuner. See figure 46. I then continue this process for each playing hole. While checking the note for each hole on the tuner, simply keep all holes above the hole being checked covered, while all holes below are open. There is one exception. While tuning the forth hole from the bottom, (which is the hole that you normally keep covered to play a Mode 1 scale), you cover the two holes above and the hole below. Then proceed in the usual manner for the remaining two holes. Don't ask me why this hole is tuned this way because I do not know. Just do it!

I mentioned using burning sticks to tune the flute. I like to burn the holes in for several reasons. First, my burning sticks are a set of drills I got on E-Bay. I made handles from pieces of dowels and rounded off the ends on a grinder. They are in 1/64 increments and you can go slow as you increase each hole size. Second, I like my playing hole to be nice and round which the drill gives me, and third, burning the holes hardens the wood around the hole and makes a nice clean job. Figure 47 shows my burning sticks.



Fig. 46

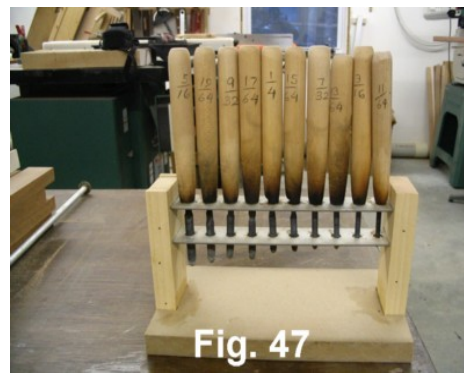


Fig. 47

Finish sanding the flute.

Now that the flute is all tuned, it is time to finish sand it. I do this on an inflatable balloon sander. There are three separate operations to sand the flute. First, I sand the area around the nest area that was not sanded on the lathe. This is done with 120 grit paper, and then sand the entire flute with 120 grit paper. Second, I sand the entire flute with 300 grit paper and third, I polish the entire flute with 500 grit paper that I have cut the grit down to nearly

nothing. This gives a very smooth polish to the flute. See figure 48



Fig. 48

Finishing the flute.

To achieve a high polish finish, I apply three to four coats of shellac to the flute. See figure 49. The first coat is a thin coat, applied with a rag and dries almost immediately. Then, after sanding the flute lightly with 600 grit paper I apply a second coat and allow it to dry for about thirty minutes. Next, I again sand the flute with 600 grit paper and apply the third coat. Again I allow it to dry for about thirty minutes. Once the flute is thoroughly dry, I add the finishing touch by rubbing the flute down with pumice and paraffin oil. This dulls the finish but makes the flute very smooth. Finally, I rub the flute with rotten stone and paraffin oil and buff it with a rag until it become glossy. Figures 50 and 51 show this last process.



Fig. 49



Fig. 50



Fig. 51

Making the bird or fetish.

The birds or fetishes I use are quite simple. However, if you desire, you can make them as elaborate as you wish. The bird should be finished in the same manner as the flute. Figure 52 shows the style of bird that I use most of the time. I save all sorts of scrap wood for this purpose.

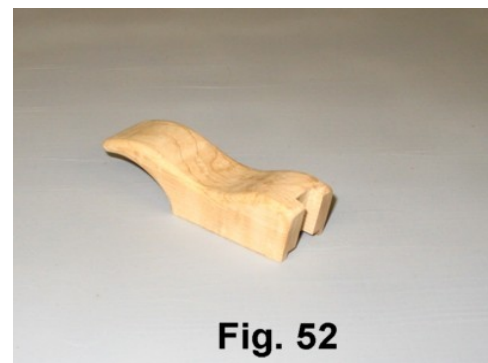


Fig. 52

Cutting the leather thong.

I usually use deer or elk hide for the thong. Elk medicine is considered to be love medicine by the Native people, and since the flute was used by the young men of the various tribes as a courting instrument, I feel it is appropriate. Any type of strong leather can be used. Again, keeping it simple, I cut a single $\frac{1}{4}$ inch to $\frac{1}{2}$ inch thong for my flutes. You can get as elaborate as you wish. There are a number of tools available to cut leather. I have two which work quite well for my needs. Figures 53 and 54 show the tools and how I cut my thong.



Fig. 53



Fig. 54

The finished flute.

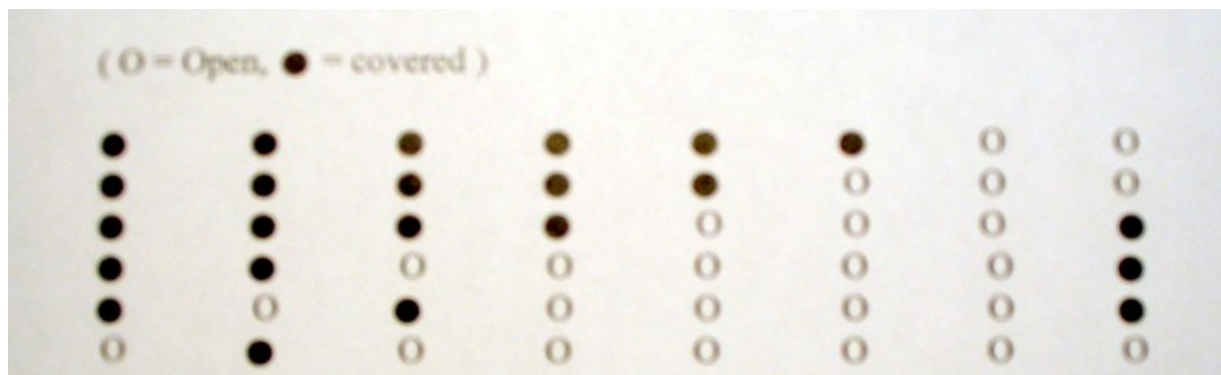
Pictured below is a finished flute. I hope you have enjoyed this booklet and if you are using it as a “how to” book, whether you are making just one flute or wish to become a flute maker, welcome to the wonderful world of the Native American Style flute, and feel free to contact me via e-mail if you have questions.



A bit more on playing the flute for beginners.

Following are a few tips for people just beginning to play the flute. Regardless of how carefully each flute is made, each will have its own personality. This is caused by differences in wood density and variations by the maker. This type of flute plays a pentatonic scale which lends itself to playing from ones heart rather than trying to play written songs, however both type of playing are possible. My flutes are tuned to both mode 1 and mode 4 of the pentatonic scale. When playing, by keeping the fourth hole from the bottom of the flute covered at all times, you will play a mode 1 scale. By keeping the third hole from the bottom covered at all times, you will play a mode 4 scale.

Once you become accustomed to playing, you may want to attempt to play written songs. This is best accomplished by learning a diatonic major scale. Below are the finger placements for an approximate major, diatonic scale.



With a bit of practice, you will soon be playing both pentatonic and diatonic scales which will open endless opportunities for you with your flute.

A few other suggestions may prove helpful.

- ☐ **Blow with a gentle breath, especially on the lower notes. Too much breath causes the notes to sound an octave higher or even to have a squeaky sound. This is called “over blowing” and may be used to create interesting effects once you have increased your proficiency.**
- ☐ **Cover the playing holes completely with the pads of your fingers rather than the tips. Again, once you advance, you can slide your fingers and partially cover certain notes . To create special effects.**
- ☐ **Occasionally the moisture from your breath may cause the flute to lose sound by moisture gathering in the airway under the bird. This is easily corrected by a sharp blast of air through the flute or by removing the bird and wiping the airway with a lint free cloth.**
- ☐ **Keep the bird tight and properly positioned. On my flutes the best position is to align the forward edge of the bird with the back edge of the TSH. Then you might have to move it slightly back and forth to get the sweet spot.**
- ☐ **Experiment. Create vibrato by pulsing your breath. Flutter your tongue and use your tongue to start notes. Create a series of sharp, rapid notes by double tonguing. Use over blowing and sliding your fingers on various holes to create special effects. Try anything you can think of. Play your flute often and carry it with you. Most of all, enjoy it and HAVE FUN!**

Appendix

Following are sources for various tools and supplies that I use in making flutes.

Carving knives (gouges)

Wood Craft Supply, www.woodcraft.com 1 800 225-1153

Wood Carvers Supply Inc., www.woodcarverssupply.com 1 800 284-6229

Files (Pillar files)

MLSC Supply, www.mlscdirect.com 1 800 645-7270

Finishing supplies (pumice, rotten stone and paraffin oil)

Wood Workers Supply Inc., www.woodworkers.com 1 800 645-9292

Lathe tools

Feather Ridge Flutes, www.featherridgeflutes.com 319 455-2562

Mortise centering jig

Eagle America, www.EagleAmerica.com 1 800 872-2511

Router cutters

MLS Inc. (free shipping), www.mlcswoodworking.com 1 800 533-9298

Sand paper

Klingspor Supply Inc., www.woodworkingshop.com 1 800 228-0000

Scorps and bent knives

North Bay Forge Inc., www.northbayforge.com (no phone number available)

Pentatonic Scales for Mode 1 and Mode 4 (Six hole flutes)

Key (Fundamental)	1st Hole	2nd Hole	3rd Hole	4th Hole	5th Hole	6th Hole
E	G	A	B	C	D	E
F	G#	A#	C	C#	D#	F
F#	A	B	C#	D	E	F#

G	A#	C	D	D#	F	G
G#	B	C#	D#	E	F#	G#
A	C	D	E	F	G	A
A#	C#	D#	F	F#	G#	A#
B	D	E	F#	G	A	B
C	D#	F	G	G#	A#	C
C#	E	F#	G#	A	B	C#
D	F	G	A	A#	C	D
D#	F#	G#	A#	B	C#	D#

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